Trine Rask Nielsen (2021): Confronting Asylum Decision-making through Prototyping Sensemaking of Data and Participation In: Proceedings of the 19th European Conference on Computer-Supported Cooperative Work: The International Venue on Practice-centred Computing on the Design of Cooperation Technologies, Reports of the European Society for Socially Embedded Technologies (ISSN 2510-2591), DOI: 10.18420/ecscw2021_dc006

Confronting Asylum Decision-making through Prototyping Sensemaking of Data and Participation

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Abstract. Despite decades of legal harmonization, the chance of receiving asylum varies significantly across Europe among people from the same country of origin. The research outlined in this paper is part of an interdisciplinary research project focused on understanding asylum decision-making and outcome variations across the Nordic countries (Sweden, Norway, and Denmark). The project will eventually cover over 100,000 asylum decisions. The research outlined in this paper aims to push agendas on data science in Computer-Supported Cooperative Work (CSCW) through prototyping context and participation as part of raising questions to the data and promoting data literacy. Combining the application of Machine Learning (ML) and Natural Language Processing (NLP) with participatory methods (e.g. critically designed artefacts) enable us to 1) move beyond "obvious" ML-application areas, 2) through sensemaking of data with stakeholders, and 3) co-develop approaches to data science from a CSCW-perspective.

Introduction

In Computer-Supported Cooperative Work (CSCW) and related areas of research scholars are starting to investigate how to respond to and create a shared research agenda for the refugee crisis (Talhouk et al. 2016, 2019). Increasing interest in data-driven technologies for decision-support has led to scholars asking new questions

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about the role of technology in relation to large-scale political issues (Dombrowski et al. 2016, Molnar 2019). As such, scholars begin to discuss the entrenched inequities of data science tools among the already marginalized and how they locate wealth and power largely in Western societies (Taylor et al. 2021). Acknowledging these deep issues of inequity "data are part of the problem, to be sure. But they are also part of the solution" (D'Ignazio et al. 2020).

The PhD research outlined in this paper is part of the interdisciplinary research project Data Science for Asylum Legal Landscaping (DATA4ALL) focused on understanding asylum decision-making and outcome variations across the Nordic countries (Sweden, Norway, and Denmark). This project will use Natural Language Processing (NLP) and machine learning (ML) together with participatory design methods to understand outcome variations in asylum decisions across Nordic countries. Despite decades of legal harmonization, the chance of receiving asylum for displaced people from the same country varies significantly across Europe (Goodwin-Gill et al.; Guild 2016).

According to the 1951 Refugee Convention Article 1A, a refugee is defined as a person:

> "who is outside his or her country of nationality or habitual residence; has a well-founded fear of being persecuted because of his or her race, religion, nationality, membership of a particular social group or political opinion; and is unable or unwilling to avail him – or herself of the protection of that country, or to return there, for fear of persecution" (The UN Refugee Agency).

The 1951 Refugee Convention does not define how states should determine refugee status. Asylum proceedings and status determinations are left to each state. Refugee status is declaratory, which means that a person becomes a refugee when they meet the criteria of the Convention, not only when a state recognizes a person as such. An asylum seeker (in this paper, described as a person seeking asylum), on the other hand, is an individual whose request for protection has not yet been processed (The UN Refugee Agency).

In this context, asylum claims are left for adjudicators to decide on the basis of testimonies and the disclosed material, which become adjudications dataset available for data science.

The research outlined in this paper contributes an approach and initial points for discussion of how a CSCW-perspective can help raise questions about large-scale data. Data from asylum decisions stems from a complex and cooperative decision-making practice, which is only opaquely described as they are extracted and made available for data-driven technologies. Contextual factors may affect the data production, including imprecise language interpretation, lacking trust in authorities – but also implicit bias of adjudicators can shape interpretations of asylum claims.

This PhD project will leverage methods for prototyping sensemaking of data and participation such as speculative and critical design (SCD) methodologies to 1) engage with different types of stakeholders, connect data to the world it came from,

and promote data literacy and ethics, and 2) advance the work of unpacking and better understanding the different systemic issues influencing variations in asylum outcomes.

Machine Learning of Asylum Decision-Making

ML is increasingly being promoted to fulfill different purposes. The DATA4ALL project will use data science techniques such as ML and NLP for the purpose of explanatory research. In other countries ML is increasingly being endorsed to support decision-making in adjudications involving foreign citizens' applications for asylum (Chen et al. 2017) in order to avoid variations in decision outcome (Ramji-Nogales et al. 2007), improve impartiality, and decrease unfair decisions made by human judges (Heyes et al. 2019; Chen et al. 2017).

Large-scale datasets are not yet being applied to asylum-decision making in Denmark, though ML has been increasingly applied in sensitive settings (e.g. social welfare) fueling debates on its use. The Danish government introduced a new National strategy in 2019 with the aim of becoming a frontrunner in AI through an ethical and responsible perspective on ML use in public casework: "The public sector should take advantage of AI to provide a world-class service" (Ministry of Industry, Business and Financial Affairs 2019).

Meanwhile, large-scale data are gaining attention in the asylum domain. In 2020 it became public that decisions of recognition rates of one judge of the Danish Refugees Appeals Board varied significantly from the decisions of other judges between 2012-2019 (Flygtningenævnet 2020). Prior CSCW-research have forcefully documented the importance of unpacking the situated and cooperative aspects of decision-making in practice. Careful investigation of the social organization of work is critical for giving context to such data (Randall et al. 2007, Møller et al. 2020).

As large-scale datasets become available, and with these a growing anticipation that they can be transformed into knowledge and informed decisions, algorithmic authority increases (Lustig et al. 2016). Legal decision-making more broadly has seen an increase in algorithmic systems for decision-support (Olsen et al. 2020). In the area of asylum decision-making, data and algorithms are increasingly considered for streamlining and adding transparency into practices around decision-making (Molnar 2019; Pakzad 2019).

If data science methods are to be applied to advance asylum law, critical questions must be raised to the data in order to ensure social and ethical aspects and legal liability related to this project. This means engaging with various stakeholders to certify that context is considered appropriately (Neff et al. 2017).

Prototyping Sensemaking of Data and Participation

This PhD project will be approached by opening a conversation between CSCW, critical data studies (Neff et al. 2017), and principles from "Data Feminism" (D'Ignazio et al. 2020). To contextualize and ground the findings from applying e.g., ML to asylum cases, I intend to use participatory methods as an approach, for example, prototyping as a strategy for enabling sensemaking of data and participation (Neff et al. 2017).

Prototyping as a method covers a spectrum from product development (Andersen et al. 2017) to mere speculative research artefacts that can form a tool for inquiry (Auger 2013; Baumer 2017; Baumer et al. 2018). Prototyping is especially useful to 1) engage stakeholders and connect large-scale data to the situated context they are produced in while promoting data literacy of stakeholders. Further, prototyping allows me to 2) unpack the different political struggles influencing variations in asylum outcomes that cannot be ascribed to the situated practices of asylum decision-making (*following* Cakici et al. 2020). 3) Data are transformative of future practices and prototyping enables stakeholders who are not trained in data sciences to co-develop methods and take part in the discussion- and design of the databased services (Seidelin et al. 2020).

To unpack asylum decision-making and understanding outcome variations from a CSCW and critical data perspective, this PhD project revolves around four research questions as described below including further elaboration of the methodological approach, descriptions of the work/findings to date, next steps and expected contributions.

RQ1: How can perspectives from the people applying for asylum be taken into consideration throughout the project?

To answer this research question, I must unpack and understand *who* the people seeking asylum are and *how* and by *whom* the people seeking asylum become categorized into "asylum seekers".

Feminist theory encourages us to seek and hear the voices of the persons who are being described or discussed. Designing data-driven technologies responsibly involves that we include the people that are eventually influenced (D'Ignazio et al. 2020). Thus, I plan to examine what a responsible approach to knowledge production looks like from the perspectives of the people seeking asylum to rebalance the perspective of government stakeholders, which produced the decisions analyzed in the project. I intend to set up inclusive practices that allows the perspectives of the people seeking asylum to be represented.

RQ2: How are asylum decisions made in Denmark today?

In order to get a deep understanding of the Danish asylum decision-making process, an overview of the current Danish asylum procedure was created (Figure 1) as a starting point for prototyping a critically designed artefact.

This overview will be further developed to engage stakeholders in a discussion of the work practices of asylum decision-making, such as the subtle categorization of cases (Møller et al. 2012).

I will use prototypes as tools for collaboration with stakeholders such as NGO's, the Danish Immigration Service, the Refugee Appeals Board, and the people applying for asylum to understand the situated context and to study the classification systems, the categories, and standards in use.

Prototyping critically designed artefacts serves two purposes in this project: 1) gaining a shared understanding and common vocabulary in regard to Danish refugee terms and processes, and 2) mutual learning (Kensing et al. 2013) through participation of all stakeholders to establish a platform for sensemaking of data and discussions of the work practices of asylum decision-making, such as the subtle categorization of cases (Møller et al. 2011; Møller et al. 2020).

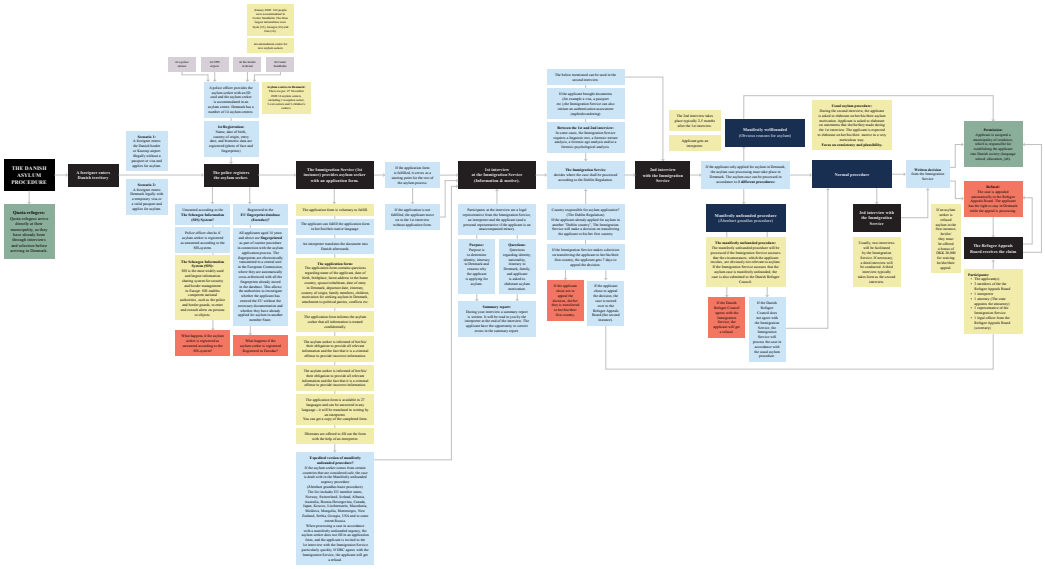


Figure 1. An overview of the current Danish asylum procedure

RQ3: What counter data can be constructed that can help nuance our insights and learnings about the Danish asylum procedure?

As a preliminary exercise in the DATA4ALL project, data from decision summaries of asylum cases treated by the Refugee Appeals Board have been extracted and analyzed. These datasets contain publicly available summaries for decision-making on asylum cases (The Refugee Appeals Board) that have been rejected by the Danish Immigration Service and therefore turned over to the Refugee Appeals Board.

The data that has informed the analysis in the preliminary exercise are - as any other data, neither neutral nor objective, but constructed and products of unequal social and political relations. When working with data, I must actively prevent

numbers from speaking for themselves and therefor situating data in context is necessary for making any real sense of correlation (Randall et al. 2007, Møller et al. 2020). It is essential to address the complexity of what the data actually represent and contextualize and evaluate the findings of this study to ensure that its "situatedness" (Haraway 1980) is taken into account.

The variations identified in the preliminary exercise will be used as a starting point for developing a second speculative research artefact. Through applying a SCD approach, it becomes possible to research “counter data” (D’Ignazion et al. 2020) through amplifying these stakeholders’ ground-up understanding of asylum-decision-making and what may cause the kinds of variations in asylum cases.

Counter data is missing data or data that are currently not being collected because of e.g., bias, lack of social and political will, or structural oppression. “Data Feminism” seeks to use data and computation to counter this kind of oppression (D’Ignazio et al. 2020). Possible questions to explore through speculative design to research counter data: What happens with data about a person seeking asylum first registered in, for example, Greece and afterwards seeking asylum in Denmark? What happens with the data when an asylum case is rejected by the Danish Immigration Service and turned over to the Refugee Appeals Board? What kind of important details could possibly get lost during these processes?

RQ4: How can values encoded and reproduced in the asylum classification system be challenged through speculative research artefacts?
Acknowledging that asylum decision-making is a highly political domain, I plan to challenge the values encoded and reproduced in the Danish asylum classification system through speculative design.

Speculative research artefacts and scenarios are not meant to be implemented or used in everyday context, rather they can act as a catalyst for both sense-making and as thinking tools to question and discuss the broader impacts and consequences of technological infrastructures and the ethical implications of data science techniques (Auger 2013; Dunne et al. 2013).

Possible questions to ask through speculative research artefacts: Who are made visible and who are made invisible in the asylum classification system in Denmark? What and whose values are encoded and reproduced in Danish asylum decision-making through the classification systems, categories, and standards in use? What do these values say about the Danish asylum system?

Acknowledgements

I would like to direct a special thanks to my supervisors Naja Holten Møller, Thomas Hildebrandt, and Thomas Gammeltoft-Hansen and to my colleagues Panagiota Katsikouli, Tijs Slaats, Anna Højberg Høgenhaug, William Hamilton Byrne, Henrik Palmer Olsen, Sarah Scott Ford, Raphaële Xenidis, Irina Shklovski,

Jakob Grue Simonsen, and Baki Cakici. This PhD project is part of the DATA4ALL project, funded by the University of Copenhagen Data+ program.

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